Physics Practice Examination

Finnish Upper Secondary Syllabus (LOPS 2021), Modules FY1–FY8

Ten Problems, Weighted Towards FY5–FY8

Suggested time: 4 hours. 10 problems, 6 points each, 60 points total. Calculators and the official formula booklet (*MAOL*) are permitted. Full solutions with units are expected; numerical answers to 2–3 significant figures unless the problem asks for an exact value. Problems are *not* in order of difficulty.

Course mapping: P1 (FY1) · P2 (FY2) · P3 (FY3) · P4 (FY4) · P5, P6 (FY5) · P7 (FY6) · P8, P9 (FY7) · P10 (FY8). Problems 5–10 carry the weight of the higher courses.

Problem 1 [FY1 — Measurement, graphical models, uncertainty]

A student determines the acceleration of free fall g with a simple pendulum. The bob has radius r and hangs on a thread of negligible mass; the *effective* length of the pendulum, the quantity that enters the formula for the period, is thread length plus bob radius. Little time is available, so she makes *one* run at each of five thread lengths and obtains:

thread length l (m)	period T (s)
0.200	0.914
0.400	1.287
0.600	1.573
0.800	1.813
1.000	2.025

The periods were measured with a photogate that starts and stops the timer itself, so the random timing error is at the millisecond level. The angular amplitude was 20.0° in every run, with a reading uncertainty of $\pm 0.5^\circ$.

a) Explain why plotting T^2 against l, rather than T against l, gives a straight line, and state what theory predicts for the slope and the intercept of that line (keep r as a symbol).

b) Determine g and its uncertainty from the scatter of the points about the fitted line. Quote $g \pm \Delta g$ with a sensible number of significant figures.

c) Determine the bob radius r from the intercept, and judge whether the intercept is significantly different from zero.

d) Your value turns out to be about 1.4% below $9.81\,\mathrm{m/s^2}$. Show quantitatively that the finite angular amplitude accounts for this, and compute the corrected result. Then state, in the language of measurement error, what kind of error this is, what it did to the estimate of g, and whether it is visible in the uncertainty of part (b). Finally, say what the experimenter should have done instead, given that the photogate is already available.

Problem 2 [FY2 — Energy production, power, efficiency]

A wind farm is planned with turbines of rotor diameter $D = 120\,\mathrm{m}$ in air of density $\rho = 1.20\,\mathrm{kg/m^3}$. The wind is described by a Rayleigh distribution, i.e. it is characterised by its mean speed $\bar{v}$, here $\bar{v} = 7.5\,\mathrm{m/s}$. In such a distribution the mean of the *cube* of the speed exceeds the cube of the mean

speed: $\overline{v^3} = \frac{6}{\pi}\,\bar{v}^{\,3}$. Each turbine delivers a rated electrical power $P_{\text{rated}} = 2.50\,\text{MW}$ above a certain wind speed, and the farm's annual capacity factor is 0.35. Finnish electricity consumption is about 90 TWh per year, and coal condensing power emits about 0.90 kg of CO_2 per kWh.

a) Derive the power flowing through the rotor disc, $P_w = \frac{1}{2}\rho A v^3$, from dimensional and physical reasoning, and evaluate it at $v = 12\,\text{m/s}$. Why can no turbine extract all of it? State the Betz limit and compare it with the fraction your turbine actually achieves at 12 m/s.

b) A turbine's average output is $\overline{P} = \frac{1}{2}\rho A\,\overline{v^3}\,C_p$, where C_p is the power coefficient. Compute first $\frac{1}{2}\rho A\bar{v}^{\,3}$, then the correct average available power, and hence C_p. Check that C_p is below the Betz limit, and explain in one sentence why using $\bar{v}^{\,3}$ instead of $\overline{v^3}$ here would be an error, and in which direction.

c) Compute the annual electrical energy of one turbine in GWh, the number of turbines needed for 1.0% of Finnish consumption, and the CO_2 avoided per turbine-year relative to coal power.

d) On one site the wind alternates between 5 m/s and 10 m/s, half the time each, so the mean speed is 7.5 m/s. Show that this site yields more average power than a site with a perfectly steady 7.5 m/s wind, and give the ratio. Then explain why a real turbine nevertheless does not deliver its maximum power at the highest wind speeds.

Problem 3 [FY3 — Thermodynamics, ideal gas, heat]

A hot-air balloon has envelope volume $V = 2.2 \cdot 10^3\,\text{m}^3$. The air outside is at $T_0 = 20.0\,^\circ\text{C}$ and pressure $p_0 = 101.3\,\text{kPa}$; the molar mass of air is $M = 29.0\,\text{g/mol}$. The envelope material has mass 180 kg; basket and burner together have mass 90 kg. The pressure inside equals the pressure outside at the open mouth, so the internal density follows from the ideal gas law. Take $c_p = 1.005\,\text{kJ/(kg K)}$ for air and $R = 8.314\,\text{J/(mol K)}$.

a) Compute the density of the outside air.

b) What must the average temperature of the air inside the balloon be for the balloon to carry a payload of 350 kg?

c) The fabric is rated for at most 120 °C. What is the largest payload the balloon can lift?

d) In steady flight the burner heats air entering through the mouth at $0.45\,\text{m}^3/\text{s}$ (measured at the internal temperature) and expels it at the top. Estimate the burner power needed to hold the temperature found in part (b). State the main approximation, and say qualitatively why the true figure is somewhat larger.

Problem 4 [FY4 — Forces, Newton's laws, energy, momentum]

Two blocks are connected by a light, inextensible string over a frictionless ideal pulley. Block A, $m_A = 4.00\,\text{kg}$, slides on a plane inclined at $\theta = 30.0^\circ$; block B, $m_B = 3.00\,\text{kg}$, hangs vertically. The coefficients of friction between A and the plane are $\mu_s = 0.25$ and $\mu_k = 0.20$. The system is released from rest, and B falls a distance $h = 0.80\,\text{m}$ before hitting the floor.

a) Show that the system really does start to move as described, with block B going down, and compute the acceleration of the blocks and the tension in the string while B falls.

b) Compute the speed with which B strikes the floor, and verify the answer with an energy balance for the whole system, listing every term.

c) After B has landed the string goes slack. Determine how much further block A slides up the plane before stopping, and whether A then stays put. Justify your answer with the numbers.

d) A now slides back down. Show that it returns to the position it had at the instant B landed, and find its speed there. At that instant the string snaps taut, jerking B (still lying on the floor) into motion. Determine the common speed of the blocks immediately afterwards and the impulse transmitted by the string, and show that the fraction of kinetic energy lost is $m_B/(m_A + m_B)$. Then determine what happens in the next instant, and explain why the system does not simply continue to move together.

Problem 5 [FY5 — Circular motion, gravitation, oscillations]

The International Space Station orbits at an altitude of 400 km. Use $M_E = 5.97 \cdot 10^{24}\,\text{kg}$, $R_E = 6.371 \cdot 10^{6}\,\text{m}$ and $G = 6.674 \cdot 10^{-11}\,\text{N m}^2/\text{kg}^2$. Ignore air drag except where asked.

a) Compute the orbital speed, the orbital period, and the local gravitational field strength at the station's altitude.

b) Show that for a circular orbit of radius r the kinetic energy of a body of mass m equals $-\frac{1}{2}U$, where U is its gravitational potential energy with $U \to 0$ at infinity. The station has mass $4.2 \cdot 10^{5}\,\text{kg}$; compute its total mechanical energy. Then compute the escape speed from 400 km and explain, using $K = -\frac{1}{2}U$, why it exceeds the orbital speed by exactly $\sqrt{2}$.

c) How much work must be done *on* the station to lower it from 400 km to 399 km? Hence estimate the changes in v and in the period, and explain the sign of the change in v — in particular why a decelerating manoeuvre makes the station go *faster*.

d) A crew member in a planned rotating habitat of angular speed $\omega = 0.256\,\text{rad/s}$ sleeps with head towards the rotation axis, feet against the outer wall at radius $R_s = 50.0\,\text{m}$. Compute the artificial gravity at the feet, and show that it is 3.6% weaker at the head of a person 1.80 m tall. Why is this gradient a problem, and what does it imply about the size of a healthy rotating habitat?

e) A 3.00 m pendulum hangs in that habitat. What is its small-oscillation period, and how does it compare with the same pendulum on Earth?

Problem 6 [FY5 — Sound, resonance, superposition]

A tuning fork marked 440 Hz produces sound whose displacement amplitude at a nearby point is 0.500 mm; the speed of sound in air is 343 m/s. In a resonance-tube experiment with a closed pipe (one open end) the first two resonances of this fork occur at effective pipe lengths $l_1 = 0.182\,\text{m}$ and $l_2 = 0.572\,\text{m}$.

a) Compute the maximum speed and maximum acceleration of the air molecules, and the wavelength of the sound. Show that $\lambda = 2(l_2 - l_1)$, and compute the speed of sound from the measurement.

b) The first resonance is found at a tube length *smaller* than the $\lambda/4$ one would predict for an ideal pipe. Explain the physical reason and determine the end correction Δl. Using $\Delta l \approx 0.6\,r$ for a flanged open end, estimate the tube radius. Why does Δl cancel out of $\lambda = 2(l_2 - l_1)$ but not out of $\Delta l = l_{\text{ideal}} - l_1$?

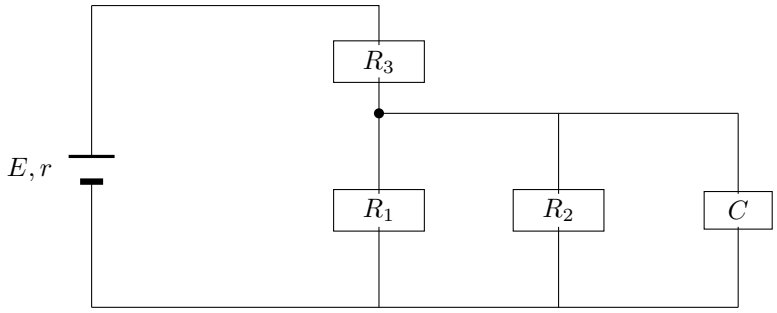
c) The same fork is sounded together with a second fork and beats of 4.0 Hz are heard. What are the possible frequencies of the second fork? Describe an experiment that decides between them, and state how many beats are heard in 5.0 s for $f_2 = 444\,\text{Hz}$.

d) The experiment is repeated in a room filled with carbon dioxide and the first resonance is then at $l_1 = 0.140\,\text{m}$. Assuming the same end correction, determine the speed of sound in CO_2 and compare it with the value predicted from $v = \sqrt{\gamma RT/M}$ using $\gamma_{CO_2} = 1.30$, $M_{CO_2} = 44.0\,\text{g/mol}$

and the corresponding air values $\gamma = 1.40$, $M = 29.0\,\mathrm{g/mol}$. Would the beat frequency of part (c) change in CO_2?

Problem 7 [FY6 — DC circuits and capacitors]

In the circuit below, $E = 12.0\,\mathrm{V}$, the battery's internal resistance is $r = 3.0\,\Omega$, $R_1 = 18\,\Omega$, $R_2 = 22\,\Omega$, $R_3 = 15\,\Omega$ and $C = 2.2\,\mu\mathrm{F}$. The capacitor has negligible leakage, and the circuit has been connected for a long time.



The pair $R_1 \parallel R_2$ is in series with R_3 and the internal resistance; the capacitor is in parallel with that pair.

a) Determine the total current, the terminal voltage of the battery, the current in each of R_1 and R_2, and the charge and energy stored in the capacitor.

b) Compute the power dissipated in every resistor, verify energy conservation, and determine the circuit's efficiency, defined as the fraction of the chemical power that is dissipated in the external resistors.

c) A voltmeter of finite resistance R_V is connected across the parallel pair. Derive how the reading depends on R_V, evaluate it for $R_V = 200\,\Omega$ and $R_V = 30\,\Omega$, and explain why these readings are systematically too small. What would you require of R_V for a 1% error?

d) An LED with forward voltage 2.0 V and rated current 20 mA is to be run from a separate 9.0 V battery of negligible internal resistance. Explain why a diode's current–voltage characteristic makes a series resistor necessary, and determine its resistance and the power it dissipates. What happens if the LED is connected the wrong way round, and why does the current fall as such a battery discharges?

Problem 8 [FY7 — Induction, generators, transformers]

a) A 100-turn coil of cross-sectional area $1.22 \cdot 10^{-2}\,\mathrm{m^2}$ rotates at 3000 revolutions per minute in a uniform field of 0.85 T. Derive the induced emf as a function of time, and compute the peak and rms values. Explain why the peak emf exceeds the average emf over a quarter turn by exactly the factor you find, and state the physical reason for the factor.

b) That generator feeds a 50 MW load. Electricity is transmitted through a line of total resistance $10\,\Omega$. Compute the line loss both without a transformer (applying the coil's own rms voltage) and with the voltage raised to 110 kV, and give the lost fraction of the power in each case. Why can a transformer not be used with a DC source? Why is the transmission voltage a question of *cost*, not of the transformer's efficiency, and what fixes the practical upper limit on it?

Problem 9 [FY7 — Induction, wave optics, optical fibre]

a) A straight metal rod of mass $m = 0.20\,\mathrm{kg}$ and length $L = 0.30\,\mathrm{m}$ slides on two parallel, frictionless, perfectly conducting rails inclined at 30.0° to the horizontal. The rails are joined at the top by

a resistor $R = 0.30\,\Omega$, so the rod and resistor form a closed loop. A uniform magnetic field of magnitude $B = 1.2\,\mathrm{T}$ is applied perpendicular to the plane of the rails. The rod is released from rest.

i. Derive the terminal speed and compute it.

ii. At the terminal speed, compute the current and verify that the electrical power equals the rate of loss of mechanical energy.

iii. From the terminal state, an external force pushes the rod *up* the incline at a constant $2.0\,\mathrm{m/s}$. Find the required force and the current, and account for the power balance among the three energy channels.

iv. Now suppose the field is *vertical* (magnitude $1.2\,\mathrm{T}$) while the rails still lie in the inclined plane. Explain why a current *does* flow (compute $\mathcal{E} = BLv\cos\theta$), and show that the terminal speed is larger by the factor $1/\cos^2\theta$ than in the perpendicular case. Why does the magnetic force now need a component along the rails of $mg\sin\theta$, and where does the extra factor of $1/\cos\theta$ come from?

b) Light of vacuum wavelength $\lambda = 500\,\mathrm{nm}$ enters a $1.00\,\mathrm{m}$ long silica fibre whose core has index $n_c = 1.46$ and cladding $n_{cl} = 1.45$.

i. Determine the critical angle at the core–cladding boundary, and the greatest angle to the fibre *axis* at which light is still guided inside the core. Hence compute the numerical aperture $\mathrm{NA} = \sqrt{n_c^2 - n_{cl}^2}$ and the acceptance angle in air.

ii. For a ray at that limiting angle, compute the geometric path length in $1.00\,\mathrm{m}$ of fibre and the time delay per kilometre relative to the axial ray. Why does this effect limit the bit rate of a step-index fibre, and what does a graded-index core do about it?

iii. The attenuation at $500\,\mathrm{nm}$ is $0.25\,\mathrm{dB/km}$. Compute the transmitted fraction after $10\,\mathrm{km}$. Then explain why silica fibres transmit infrared far better than violet, using the idea that absorption requires a photon energy matching an electronic transition.

Problem 10 [FY8 — Quantisation, atoms, nuclei, radiation]

a) A sodium lamp emits the doublet at $589.0\,\mathrm{nm}$ and $589.6\,\mathrm{nm}$.

i. Compute the photon energies in eV, their difference in meV, and the frequency difference in GHz.

ii. The light passes through a grating with 600 lines per millimetre. Determine all diffraction maxima that exist and their angles, and how many lines must be illuminated to resolve the doublet in first order.

iii. Light of $500\,\mathrm{nm}$ falls on a metal with work function $2.28\,\mathrm{eV}$. Compute the threshold wavelength, the maximum kinetic energy of the photoelectrons and the stopping potential. Which of these quantities is independent of the light's intensity, and what does the intensity control instead?

b) The D–T fusion reaction is $^2\mathrm{H} + {}^3\mathrm{H} \to {}^4\mathrm{He} + n$, with masses 2.014102, 3.016049, 4.002602 and $1.008665\,\mathrm{u}$, where $1\,\mathrm{u} = 931.494\,\mathrm{MeV}/c^2$.

i. Compute the energy released per reaction in MeV and, taking only the deuterium as fuel, the energy density in J/kg. Compare it with U-235 fission ($\approx 200\,\mathrm{MeV}$ per nucleus) and with coal ($\approx 30\,\mathrm{MJ/kg}$).

ii. A fusion plant produces $2.5\,\mathrm{GW}$ of fusion power. How many reactions per second does that require, and what mass of deuterium per year? Assuming seawater contains about $33\,\mathrm{g}$ of

deuterium per cubic metre, estimate the volume of seawater needed per year, and judge whether fuel supply is the limiting engineering problem.

c) A ^{131}I sample of activity 20.0 GBq is prepared for a patient; the half-life is 8.02 d.

i. What activity remains after 12.0 h? Explain with $N = N_0 e^{-\lambda t}$ why the number of nuclei in the sample does not visibly change over that time, and compute the initial number of nuclei.

ii. Why is ^{131}I — a β^- and γ emitter with an eight-day half-life — well suited to thyroid imaging, whereas a pure α emitter of the same activity would be useless for it?

d) One patient absorbs 0.20 Gy from a gamma source distributed through the body; another absorbs the same 0.20 Gy from alpha emitters concentrated in lung tissue. Compute the equivalent dose in each case using $\mathrm{Sv} = \mathrm{Gy} \cdot w_R$, with $w_R = 1$ for photons and $w_R = 20$ for alpha particles, and judge which case carries the greater health risk in terms of *equivalent* dose. Then state the further factor this conversion ignores and which makes the two cases differ even more.